

Q1: choose the correct answers:

(Please write your answers in the table only; otherwise, your answers will be neglected)

Question	1	2	3	4	5	6	7	8	9	10
Answer										

(15 marks)

- The twisted pair cable in which metal casing improves the penetration of noise or crosstalk is called
 - insulated twisted pair cable
 - Shielded twisted pair cable
 - Unshielded twisted pair cable
 - Both A and B
- Multiplexing combines signals from several sources to achieve
 - Data rate management
 - Interleaving
 - TDM efficiency
 - bandwidth efficiency
- The synchronous Time-Division Multiplexing (TDM) is not efficient because of
 - higher data rate
 - infinite frames
 - empty slots
 - Both A and C
- The multiplexing technique that is used when the data rate of an input line is a multiple of the others is called
 - multilevel multiplexing
 - multiple-slot allocation
 - pulse stuffing
 - Asynchronous TDM

5. TDM is a digital multiplexing technique for combining several low-rate channels into
- A. a group of low-rate channels
 - B. one high-rate one channel
 - C. two high-rate channels
 - D. None
6. CRC stands for
- A. combine resistance check
 - B. cyclic redundancy code
 - C. combine redundancy code
 - D. cyclic redundancy check
7. There is no error in the received code word, if the syndrome is
- A. maximum
 - B. minimum
 - C. 0
 - D. 1
8. To guarantee the detection of up to s errors in all cases, the minimum hamming distance in a block code must be
- A. s
 - B. $s+1$
 - C. $s-1$
 - D. 0
9. Radio waves are
- A. unidirectional
 - B. omnidirectional
 - C. bidirectional
 - D. directional
10. Optical fibers use reflection to guide light through a
- A. channel
 - B. metal wire
 - C. light
 - D. plastic

Q2: Ten sources, six with a bit rate of 200 kbps and four with a bit rate of 400 kbps, are to be combined using multilevel TDM with no synchronizing bits. Answer the following questions about the final stage of the multiplexing:

- a. What is the size of a frame in bits?
- b. What is the frame rate?
- c. What is the duration of a frame?
- d. What is the data rate

(12 marks)

Q3: Draw the constellation diagram for the following cases. The numbers in parentheses define the values of I and Q , respectively.

- a. Two points at $(2, 0)$ and $(3, 0)$
- b. Two points at $(3, 0)$ and $(-3, 0)$
- c. Four points at $(2, 2)$, $(-2, 2)$, $(-2, -2)$, and $(2, -2)$
- d. Two points at $(0, 2)$ and $(0, -2)$

(12 marks)